

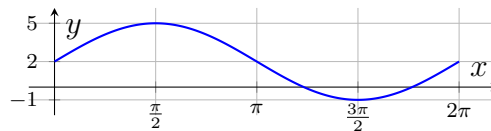
Modeling Periodic Behavior from Graphs

Reading amplitude and midline from a graph, recovering the period, writing the model, and solving with it

Directions.

- Every model here has the form $y = A \sin(Bx) + D$ or $y = A \cos(Bx) + D$, possibly with a horizontal shift.
- Read maxima, minima and periods off the graph; then compute. Exact values (fractions, multiples of π) unless a unit is asked for.
- Amplitude is measured from the *midline*, never from the horizontal axis.

1. The graph below shows one full cycle of a sinusoid.



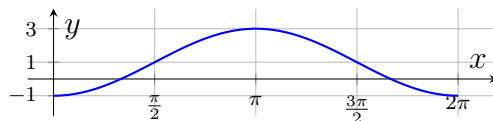
- (a) State the amplitude.

Answer: _____

- (b) State the value of D on the midline $y = D$.

Answer: _____

2. The graph below shows one full cycle of a sinusoid.



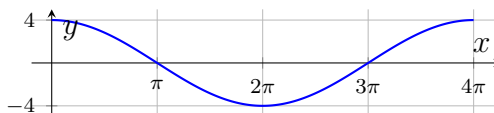
- (a) State the amplitude.

Answer: _____

- (b) State the value of D on the midline $y = D$.

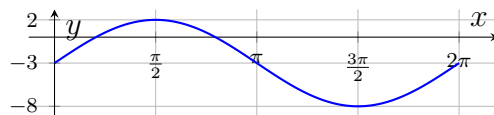
Answer: _____

3. The graph below shows exactly one cycle of $y = A \cos(Bx)$. Read the period from the graph, then find B .



Answer: _____

4. The graph below shows one full cycle of a sinusoid.



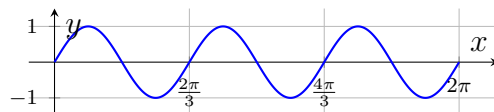
- (a) State the amplitude.

Answer: _____

- (b) State the value of D on the midline $y = D$.

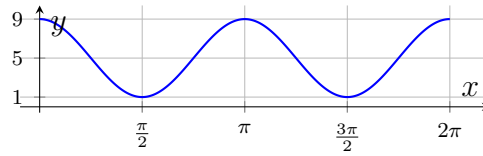
Answer: _____

5. The graph below shows $y = \sin(Bx)$. Read the period of one cycle from the graph, then find B .



Answer: _____

6. The graph below is modelled by $y = A \cos(Bx) + D$ with $A > 0$.



- (a) Find A .

Answer: _____

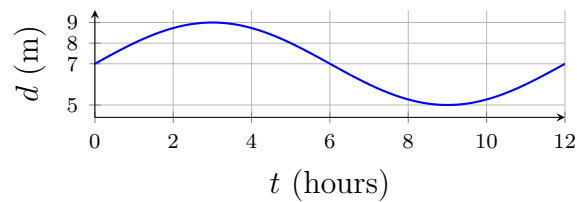
- (b) Find D .

Answer: _____

- (c) Find B .

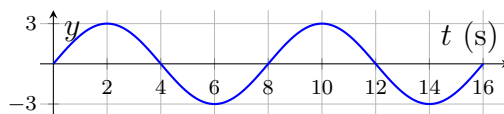
Answer: _____

7. The depth of water at a harbour mouth is modelled by $d(t) = 2 \sin\left(\frac{\pi t}{6}\right) + 7$, where d is in metres and t is the number of hours after midnight. The graph shows the first 12 hours. Find every time t with $0 \leq t < 12$ at which the depth is exactly 8 metres.



Answer: _____ hours

8. A vibration sensor records the graph below over 16 seconds. Two complete cycles are shown. The motion is modelled by $y = A \sin(Bt)$. Read the period, then find B .



Answer: _____

9. A ferris wheel has diameter 40 metres and its centre sits 22 metres above the ground. It turns once every 8 minutes. A rider boards at centre height and rises first, so the height is modelled by $h(t) = A \sin(Bt) + 22$ metres, t in minutes.

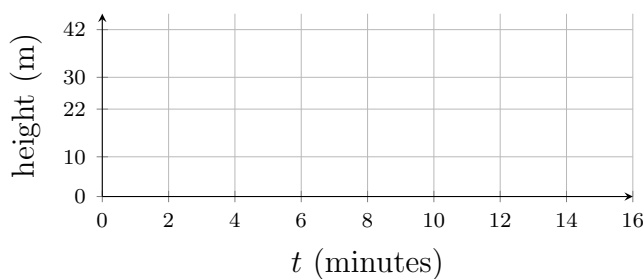
(a) Find the amplitude A .

Answer: _____ m

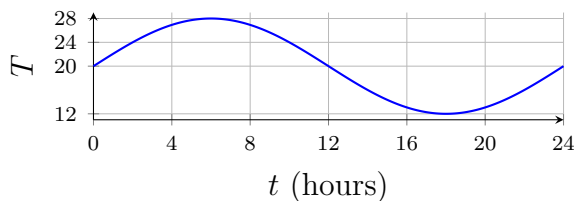
(b) Find B .

Answer: _____

- (c) Sketch two complete revolutions on the grid below. Label the midline, the maximum and the minimum.

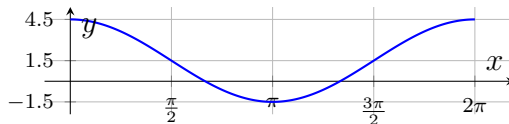


- 10.** The temperature in a greenhouse is modelled by $T(t) = 8 \sin\left(\frac{\pi t}{12}\right) + 20$, where T is in degrees Celsius and t is the number of hours after 6 a.m. Find every time t with $0 \leq t < 24$ at which the temperature is exactly 24 degrees Celsius.



Answer: _____ hours

- 11.** The graph below shows one cycle of a sinusoid whose maximum is 4.5 and whose minimum is -1.5 .



- (a) State the amplitude.

Answer: _____

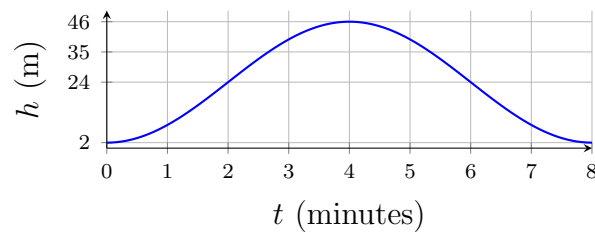
- (b) State the value of D on the midline $y = D$.

Answer: _____

- (c) Explain why the amplitude is not simply the maximum value shown.

Answer: _____

12. A rider's height on a wheel is graphed below over one full revolution. The maximum height is 46 metres, the minimum is 2 metres, the first maximum occurs at $t = 4$ minutes, and one revolution takes 8 minutes. The model has the form $h(t) = A \cos(B(t - 4)) + D$, with t in minutes.



(a) Find A .

Answer: _____ m

(b) Find D .

Answer: _____ m

(c) Find B .

Answer: _____

(d) Using your model, find every time t with $0 \leq t < 8$ at which the rider is exactly 35 metres above the ground. Give exact values.

Answer: _____ minutes